

Technical Document

Forget-lah

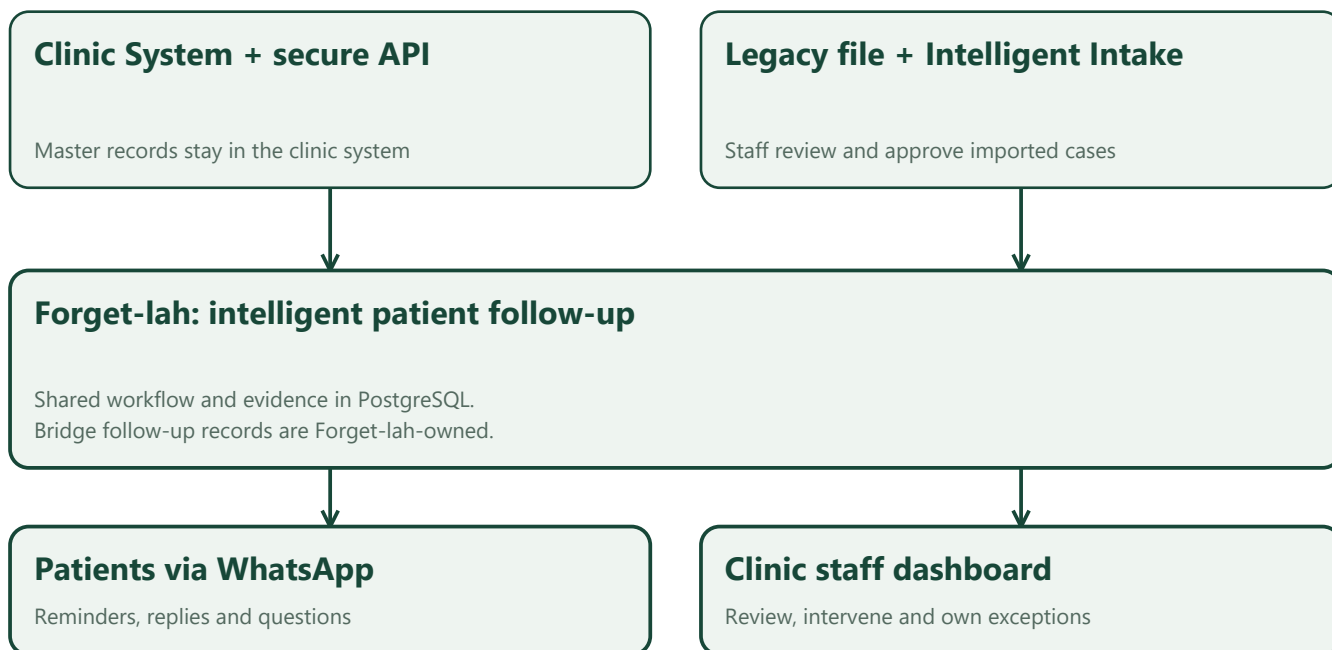
SP-ARK Agents | Team code: N63VHYEX

Evidence-backed follow-up, with bounded agentic reasoning.

Architecture, source ownership, controls, deployment and verification | 26 September 2026

System purpose and boundary

Forget-lah detects eligible follow-up, coordinates patient replies against clinic instructions, validates administrative actions and routes exceptions to staff. It complements clinic systems and clinic staff. It does not provide diagnosis, treatment decisions, a full EMR or unrestricted scheduling.



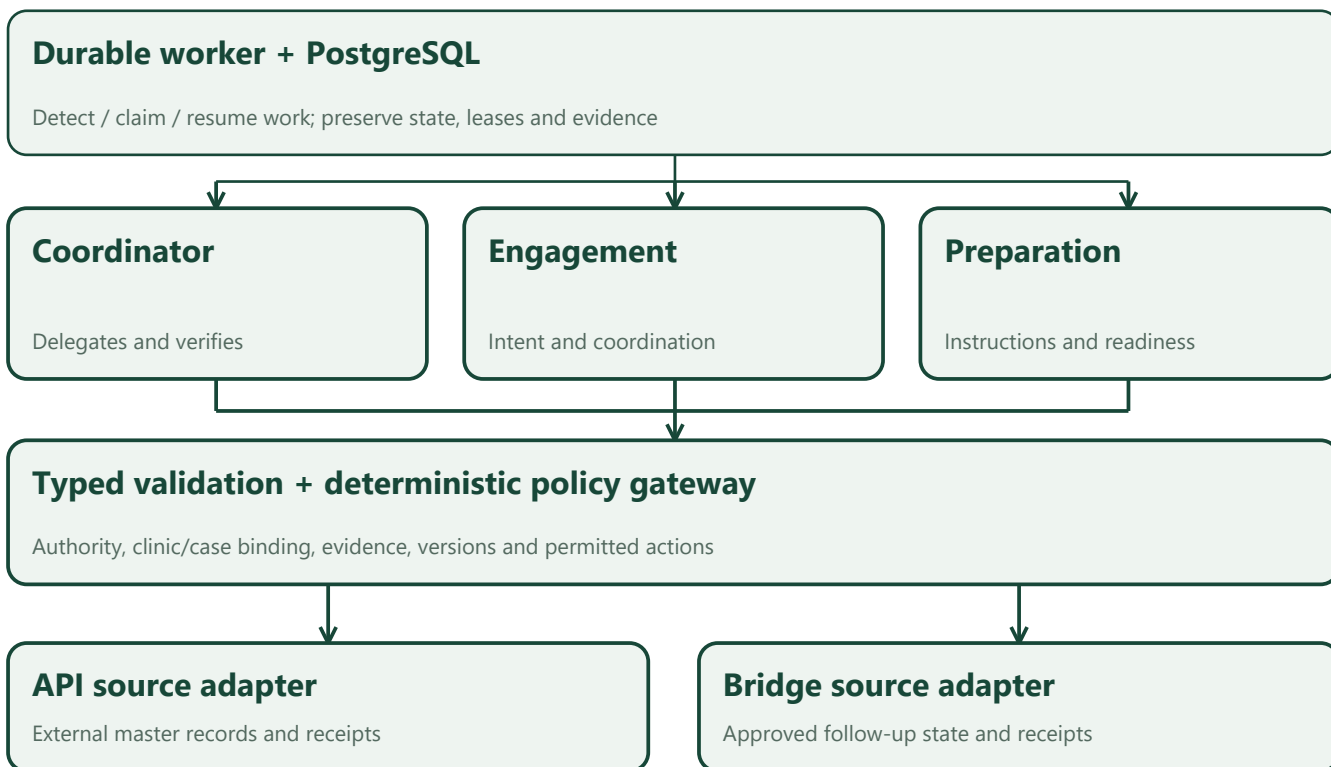
Implementation snapshot

React staff UI, FastAPI application, Python worker, PostgreSQL, source adapters, WhatsApp test channel and a model adapter using the organiser gateway. Three logical agent roles share the worker; they are not three independently deployed agent servers.

Source basis: repository HEAD 7e77ad8 plus inspected working-tree documentation. Release evidence is taken from dated repository records; it is not a new live deployment or load test. Older status passages are superseded by the latest dated implementation and release entries.

Repository: <https://github.com/satworkz/forget-lah>

Worker, agents and the control boundary



Coordinator is the only role permitted to delegate. The arrows above show the execution/control layers, not an unrestricted specialist-to-specialist delegation chain.

Role	Responsibility	Output / boundary
Coordinator	Choose bounded next steps, delegate and verify completion or handoff.	Typed decisions grounded in current case evidence; cannot bypass policy.
Engagement	Interpret reply intent, preferences and administrative coordination.	Clarification or permitted source action proposal; only source-provided options.
Preparation	Interpret doctor instructions, prerequisites and contextual conflicts.	Source-grounded findings and question coverage; unresolved clinical questions go to staff.

Execution loop

Detect and persist eligible work; claim it under a lease; load current state and authorised source evidence; resolve a clear rule or request a bounded model proposal; validate the typed result and policy; execute an allowed tool; save evidence and checkpoint. Wait for a reply, continue, or create a staff handoff. Waiting does not require an open browser or model conversation.

Inputs are treated as data, not authority. Completion requires appropriate evidence and a verified source receipt or named staff acceptance; generated wording cannot establish success.

Data ownership and integrations

Data layer	Responsibility and ownership
External Clinic System	API-backed patient and appointment master records stay external. The hackathon uses a synthetic mock-clinic service to demonstrate this boundary.
Forget-lah PostgreSQL	Persist cases, jobs, runs, decisions, delegations, replies, pending notifications, handoffs, source evidence and audit events.
Bridge lifecycle	Staff-approved imports, provenance, managed episodes and staff-supplied follow-up options belong to Forget-lah. Bridge does not write mock-clinic tables.
Model context	Per-call bounded case evidence. The model is not the memory store and has no unrestricted SQL/database access.

Intelligent Intake

CSV, TSV and XLSX exports are profiled and mapped to a bounded follow-up contract. Deterministic checks validate dates, required fields and source grounding. Uncertain rows stay in REVIEW. Staff can correct mappings/rows and approve READY records; approval creates Bridge-owned cases with provenance. Imported records remain protected from duplicate reprocessing.

Source action and message delivery are separate

A patient selection or staff action must match the relevant case, episode, option, authority and current source version. The source validates the operation and returns a receipt. Idempotent handling prevents a retry from becoming a duplicate update. A successful change with a failed notification remains committed; notification can be retried independently.

WhatsApp and after-hours use

Twilio WhatsApp Sandbox supports designated test participants and bound patient exchanges. Signed inbound handling and saved conversation context support case routing. Patient-initiated follow-up questions can be processed outside clinic hours while the service is running, but sandbox window/enrolment rules remain. Queued is not delivered. Proactive production templates, SMS and voice are not delivered capabilities.

No available source options

Unsupported availability or a source conflict triggers clarification or staff handling as appropriate. Automatic waitlisting and later slot-availability notifications are future work; the demo does not establish those capabilities.

Selective reasoning and responsible model use

Can deterministic code resolve this safely?

Use deterministic code	Use bounded model assistance
Exact supported protocol replies; saved state and transitions	Natural-language intent, ambiguity and meaningful preferences
Mandatory reads; ready validated option dispatch	Doctor-instruction interpretation and contextual conflict
Permissions, policy, option/source validation and receipts	Multilingual meaning and grounded response composition
Audit, queues and fixed failure acknowledgements	Intelligent Intake semantic mapping and normalization

All paths converge on typed validation, policy and authorised source actions. Reducing calls never removes those controls.

Organiser gateway

The dated 25 September deployment record selects Claude Sonnet 4.5 through the organiser-provided gateway for agents, intake and translations. Credentials remain server-side. Malformed or unsupported output fails validation; organiser failure does not silently switch to direct-provider or mock inference.

Cost, latency and resource controls

Agent outputs are bounded to 512 requested tokens; intake/translation uses the tested 1,024-token bound. Shared request budgets, execution-step limits, pacing, bounded retries and role-scoped context constrain consumption. Under this provider, intake uses a profile request plus one normalization request per row, so larger imports still incur more calls and time.

Recorded evidence, carefully scoped

The performance document reports a controlled offline slot-offer comparison of 7 model calls with optimisations versus 11 without, preserving the response and offered option. This is a scenario-specific call reduction, not a measured proportional reduction in live WhatsApp latency or a general cost-saving result.

Operator review

Review model attempts, errors, per-case usage, token counts where available, and provider limits alongside hosting and messaging costs. The pilot should report cost per verified outcome and failed-case overhead. No organiser account balance, expenditure or production throughput is inferred from these controls.

Security and privacy by architecture

Layer	Implemented protection
Identity and access	Hashed sessions, Argon2 password hashes, expiry, origin/CSRF checks and secure cookies with HTTPS. Staff/admin/auditor/viewer permissions are scoped per clinic.
Queued work	Worker rechecks active authority before tool execution. Case/run/clinic relationships and policy checks prevent stale authority from authorising actions.
Model boundary	Structured identifiers and contact metadata are removed from role-scoped reasoning observations. No arbitrary SQL or database credentials are available to agents.
Action boundary	Typed schemas, allowlisted tools, source evidence, case/source versions, instruction checks and deterministic policy govern actions.
Audit and provenance	Security & Audit combines policy/tool events, staff actions, receipts, notification outcomes and intake provenance, with actor/source/result where recorded.
Infrastructure	HTTPS at the cloud entry point; backend-only credentials; no published database or application API ports in the documented deployment.

Privacy: minimisation, not anonymisation

Free-text patient replies and notes can contain personal information even after structured identifiers are removed. Intake interpretation necessarily processes uploaded row content. It is therefore incorrect to claim that no model input can be related to a person. The hackathon uses synthetic records; real-patient operation requires approved purposes, vendor/data-flow assessment, retention decisions and clinic governance.

Upload safety

Bounded CSV/TSV/XLSX parsing rejects oversized or malformed input. XLSX is parsed as data, without execution or external fetching; macros, formulas, external links and unsafe XML/ZIP structures are rejected. Upload provenance includes filename metadata, actor, time and checksum.

Explicit limits

These controls do not constitute healthcare certification. Application scoping is not PostgreSQL row-level security. Audit is not cryptographically immutable. Internal container traffic is not separately encrypted. MFA, verified encryption-at-rest controls, central alerts, retention/deletion automation and tested off-host recovery are future production hardening.

Resilience, deployment and observability

Built with high availability in mind; production availability remains to be demonstrated.

Current execution design

PostgreSQL is the durable work queue. Row locks, leases, stale-result rejection, shared budgets and idempotent receipts protect execution. Separate loops serve agent work, message delivery, translation and source detection. The documented default supports two concurrent case-execution slots, configurable up to four; a single case still executes in order.

Terminal failure and staff fallback

Terminal policy, business, source or model failures create or reuse a durable staff review when authority and case bindings permit. Eligible patient replies receive a fixed, grounded failure acknowledgement without an additional model call. Repeated recovery cannot duplicate the task or acknowledgement. Manual pauses remain paused. Uncertain source writes are not blindly replayed.

A database outage can prevent persistence itself. Recovery depends on the database and supervised worker returning; immediate acknowledgement is not guaranteed. A recorded handoff still needs named staff acceptance.

Cloud deployment evidence

Repository deployment records describe a Singapore AWS Lightsail host running the web proxy, application, worker, PostgreSQL and synthetic clinic service. Containers share a single host, so the host remains a failure point. Persistent volumes protect against container replacement but are not off-host backups.

Documented demo URL: <https://forget-lah.52-220-111-31.sslip.io>

The URL is recorded deployment evidence, not a fresh availability check. Staff access requires authorised credentials; no credentials are included in this document.

Future scale and monitoring

Reactive wake-ups can use PostgreSQL notifications after durable job commits, retaining polling for recovery. Future redundant services, managed database failover and tested recovery build on the saved-state design. Capacity tests must cover concurrent cases, provider quotas, source contention, queue age and delivery delays.

Current case journeys expose saved decisions, policy verdicts, tools and outcomes. Extend this with correlated traces across inbound events, worker/model/source/delivery steps; latency and queue dashboards; alerts on stuck work and unowned handoffs; and tested retention. Traces should carry identifiers for correlation without recording unnecessary patient text or secrets.

Verification and evidence register

Evidence is reported at its recorded scope. No new deployment, patient contact or paid-model regression was performed to prepare this document.

Evidence	What it supports / limitation
Finished clinical and staff demos	Ahmad escalation; Priya prerequisite-constrained change; Alex multilingual contextual reasoning; Intake review/approval; staff change and audit. Synthetic workflows, not clinical efficacy.
Security validation record	624 passed / 14 expected skips in a recorded broad Docker/PostgreSQL run; later focused checks also recorded. Counts overlap and are not summed.
Organiser integration record	156 focused tests passed / 1 skipped; separate 33 PostgreSQL/provider tests passed. Bounded live synthetic checks recorded; not a full load test.
Performance regression	7 vs 11 model calls in one offline slot-offer comparison. No general live-latency or production-throughput claim.
Failure handling tests	Policy/source/model failure, authority and binding checks, duplicate recovery and PostgreSQL concurrent recovery coverage are documented.
Deployment verification	Dated AWS records include readiness, authenticated access, migration/release preservation and provider checks. Current uptime is not asserted.

Pilot acceptance

Test routine paths plus contradictory replies, clinical concerns, missing prerequisites, stale options, malicious instructions, tenant/role denial, malformed intake, source failure and failed delivery. Require saved evidence and correct staff ownership, not merely a plausible patient-facing message. Use synthetic records before any clinic-approved pilot.

Source register

All paths below are relative to the linked repository. They let reviewers trace the implementation without exposing deployment secrets.

README.md; docs/IMPLEMENTATION_STATUS.md; docs/FEATURE_GUIDE.md; docs/BRIDGE_INTAKE.md; docs/SECURITY_ARCHITECTURE.md; docs/PERFORMANCE.md; docs/ORGANISER_GATEWAY.md; docs/FAILURE_HANDLING.md; docs/AWS_DEMO.md; docs/CASE_JOURNEY.md.

Principal implementation areas: src/forget_lah/worker.py; runtime/engine.py; runtime/provider.py; runtime/failures.py; policy.py; security.py; auth.py; bridge.py; staff_changes.py; channel.py; apps/web/src. Verification areas include tests/test_failure_handling.py, tests/test_postgres.py and tests/test_organiser_integration.py.

Future scope

Offline Patient Companion; voice; caregiver and trusted-identity/consent integrations; follow-up pattern intelligence; diagnostic-provider slots; group reminders; reactive wake-ups and production hardening. These remain distinct from the delivered prototype.